

Supporting Information

Laser-Ablative Engineering of ZrN-based Nanoparticles for Photothermal Therapy and SERS-based Biological Imaging

Andrei I. Pastukhov[†], Victoria O. Shipunova[‡], Julia S. Babkova^{‡,¶}, Ivan V. Zelepukin^{‡,¶}, Micah Raab^{||}, Rebecca Schmitt^{||}, Ahmed Al-Kattan[†], Artem Pliss^{||}, Andrey Kuzmin^{||}, Anton A. Popov[¶], Sergey M. Klimentov[¶], Paras N. Prasad^{¶,||,*}, Sergey M. Deyev^{‡,¶,*}, Andrei V. Kabashin^{†,*}

[†] Aix-Marseille University, CNRS, LP3, Marseille, 13288, France

[‡] Shemyakin-Ovchinnikov Institute of Bioorganic Chemistry of Russian Academy of Sciences, Moscow, 117997, Russia

[¶] MEPhI, Institute of Engineering Physics for Biomedicine (PhysBio), Moscow, 115409, Russia

^{||} The State University of New York, Buffalo, Department of Chemistry and Institute for Lasers, Photonics, and Biophotonics, University at Buffalo, New York, 14260, United States

*Corresponding authors: biomem@mail.ru, pnprasad@buffalo.edu and andrei.kabashin@univ-amu.fr

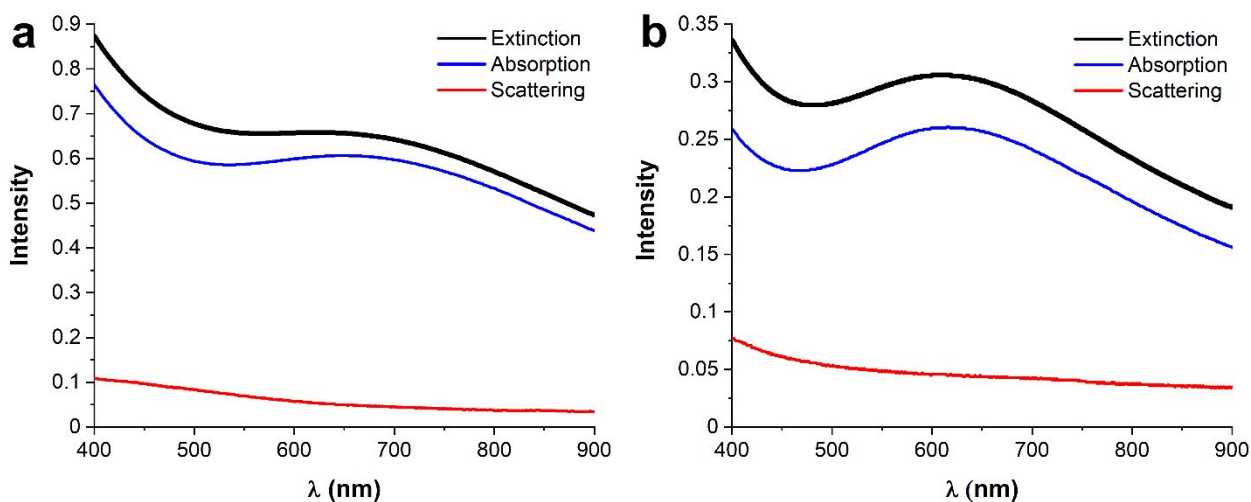


Figure S1. Optical characteristics of ZrN NPs solutions. Solutions synthesized in (a) acetone and (b) ethanol.

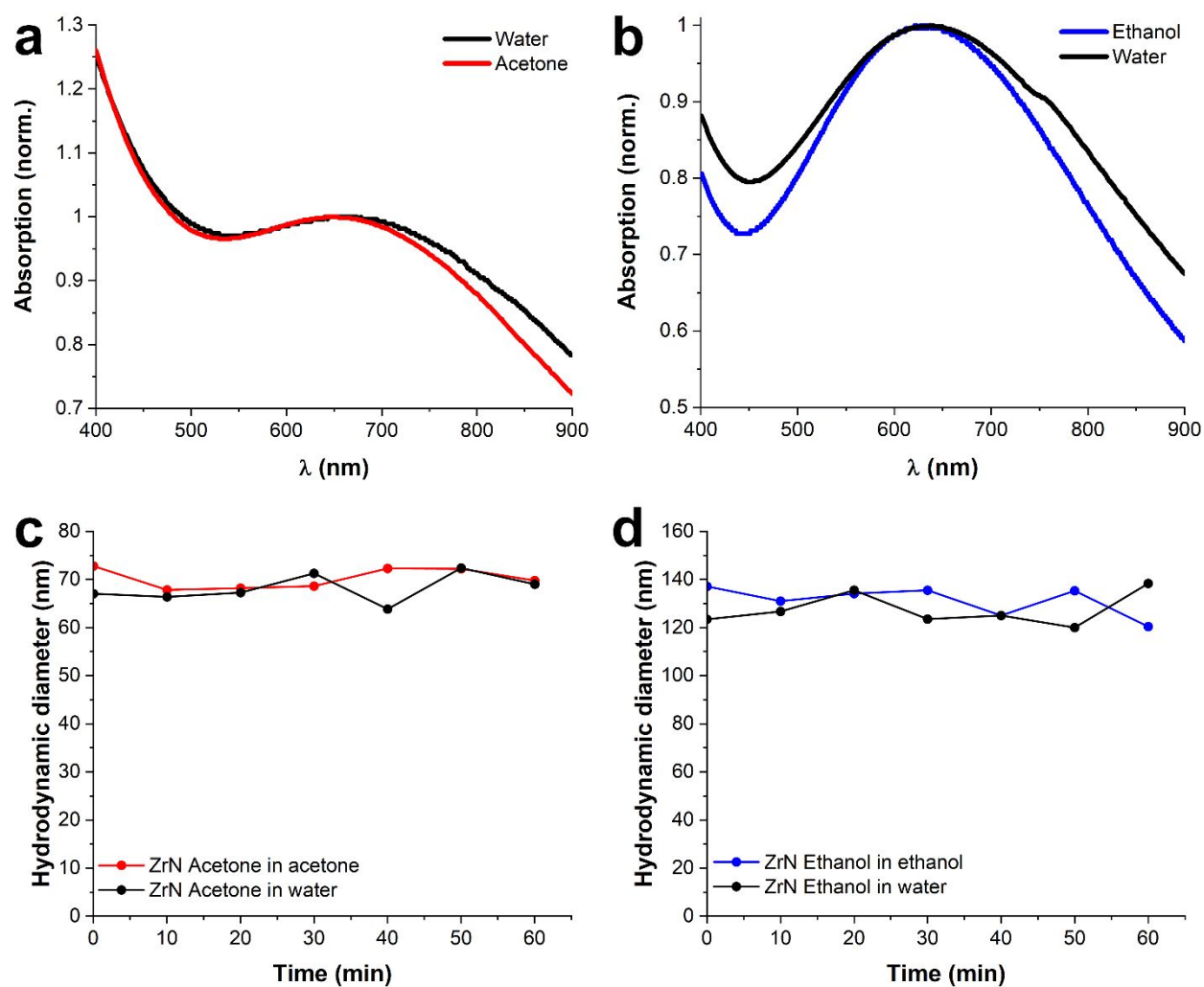


Figure S2. Normalized absorption spectra and stability evaluation of ZrN-based NPs. Samples prepared in (a,c) acetone and (b,d) ethanol before and after transfer to water.

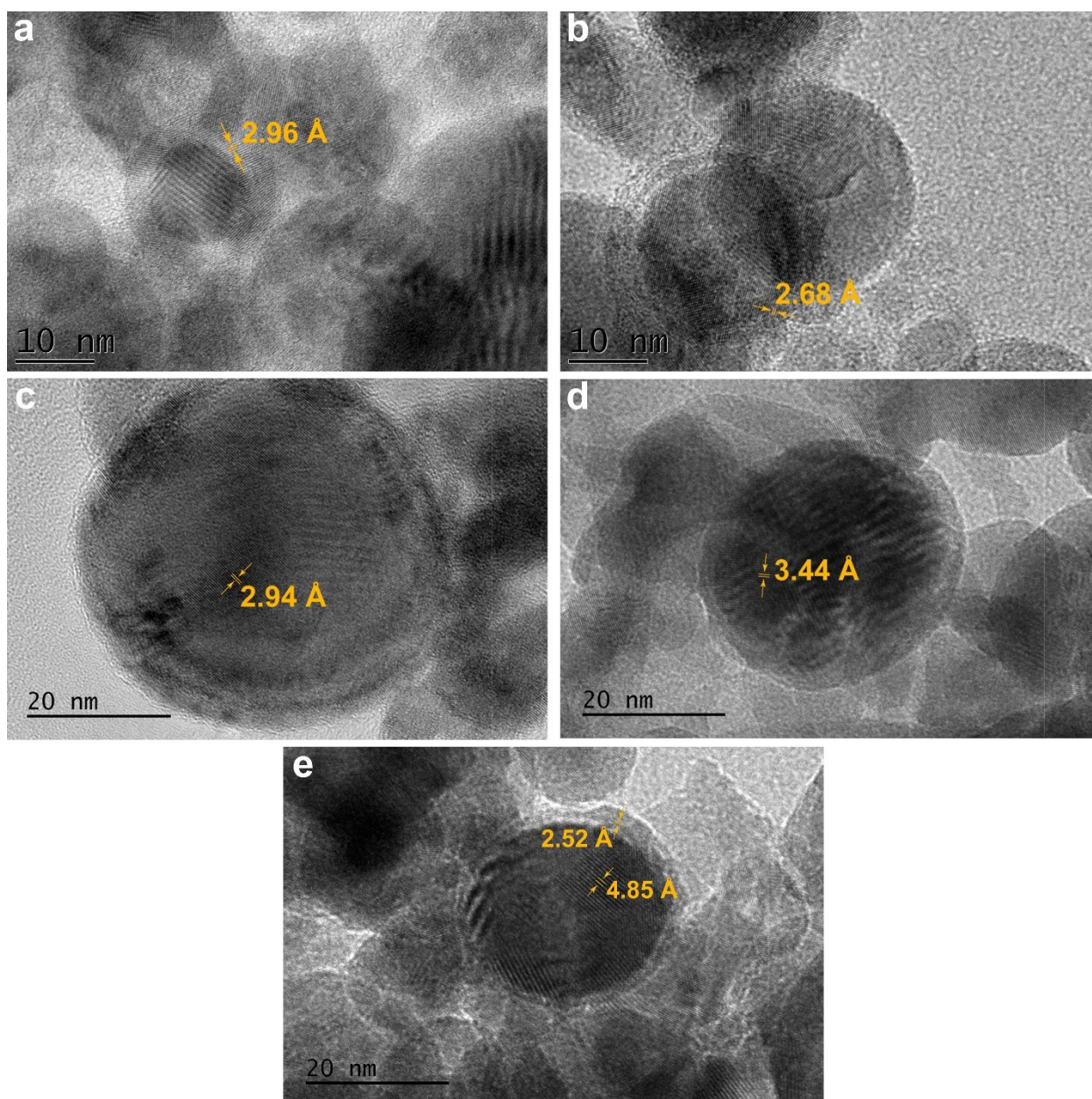


Figure S3. High resolution transmission electron microscopy (HR-TEM) images of ZrN-based core@shell nanostructures. Samples prepared in (a,b) acetone and (c-e) ethanol.

Table S1. Calculated crystallite sizes using Scherrer equation for nanoparticles prepared in acetone and ethanol

	D [nm]			
	t-ZrO ₂	m-ZrO ₂	ZrO	ZrN
Acetone	14	23	11	23
Ethanol	16	19	-	23

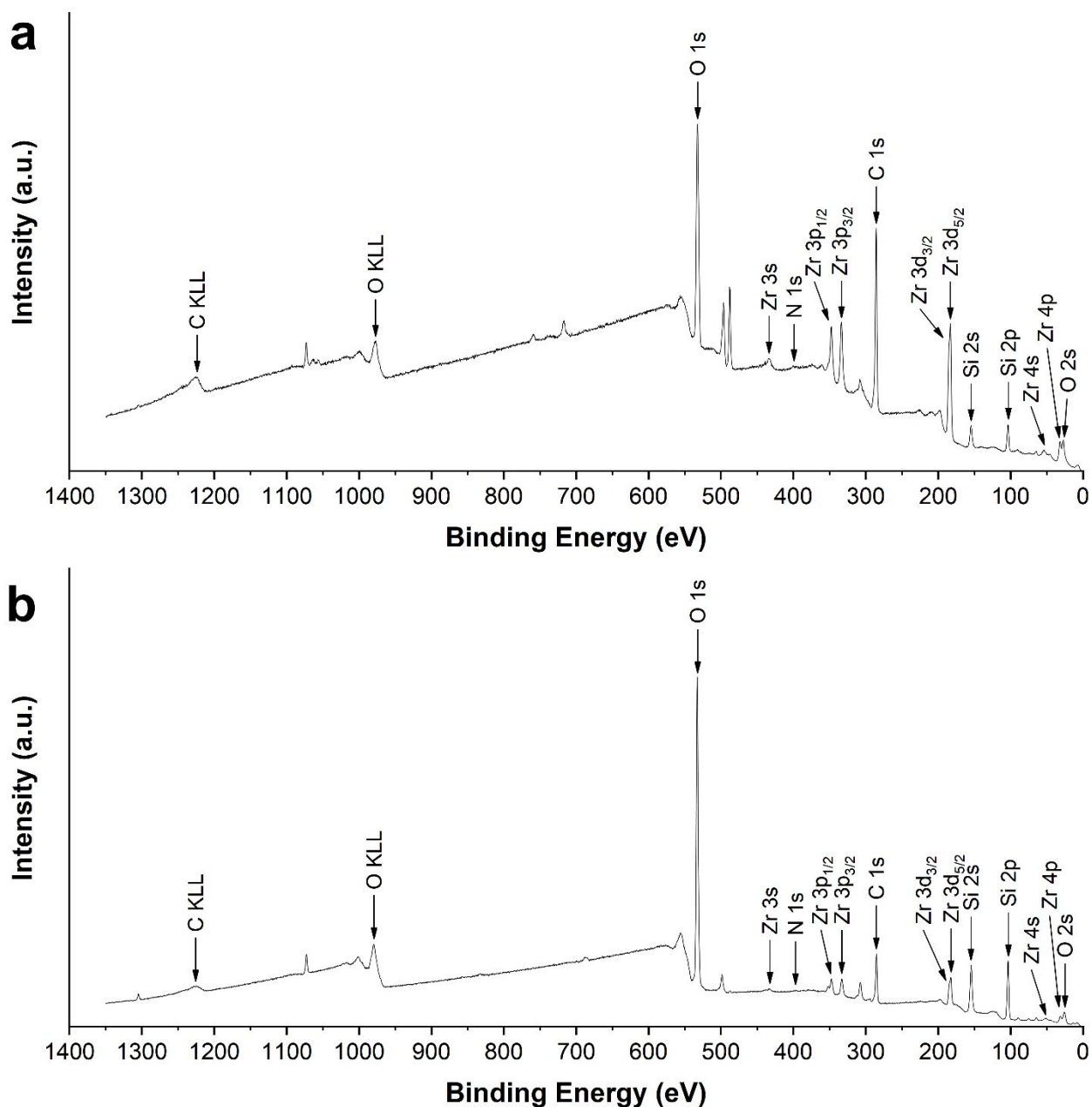


Figure S4. XPS survey spectra. Nanoparticles synthesized in (a) acetone and (b) ethanol. Silicon peaks are associated with a signal that comes from a glass substrate. Additional peaks are due to contaminants on the substrate surface.

Table S2. Atomic concentration determined by XPS analysis for nanoparticles prepared in acetone and ethanol

Solution	Atomic Concentration [%]				
	C 1s	N 1s	O 1s	Si 2p	Zr 3d
Acetone	51.65	0.64	32.52	8.58	6.61
Ethanol	20.08	0.29	51.38	26.03	2.22

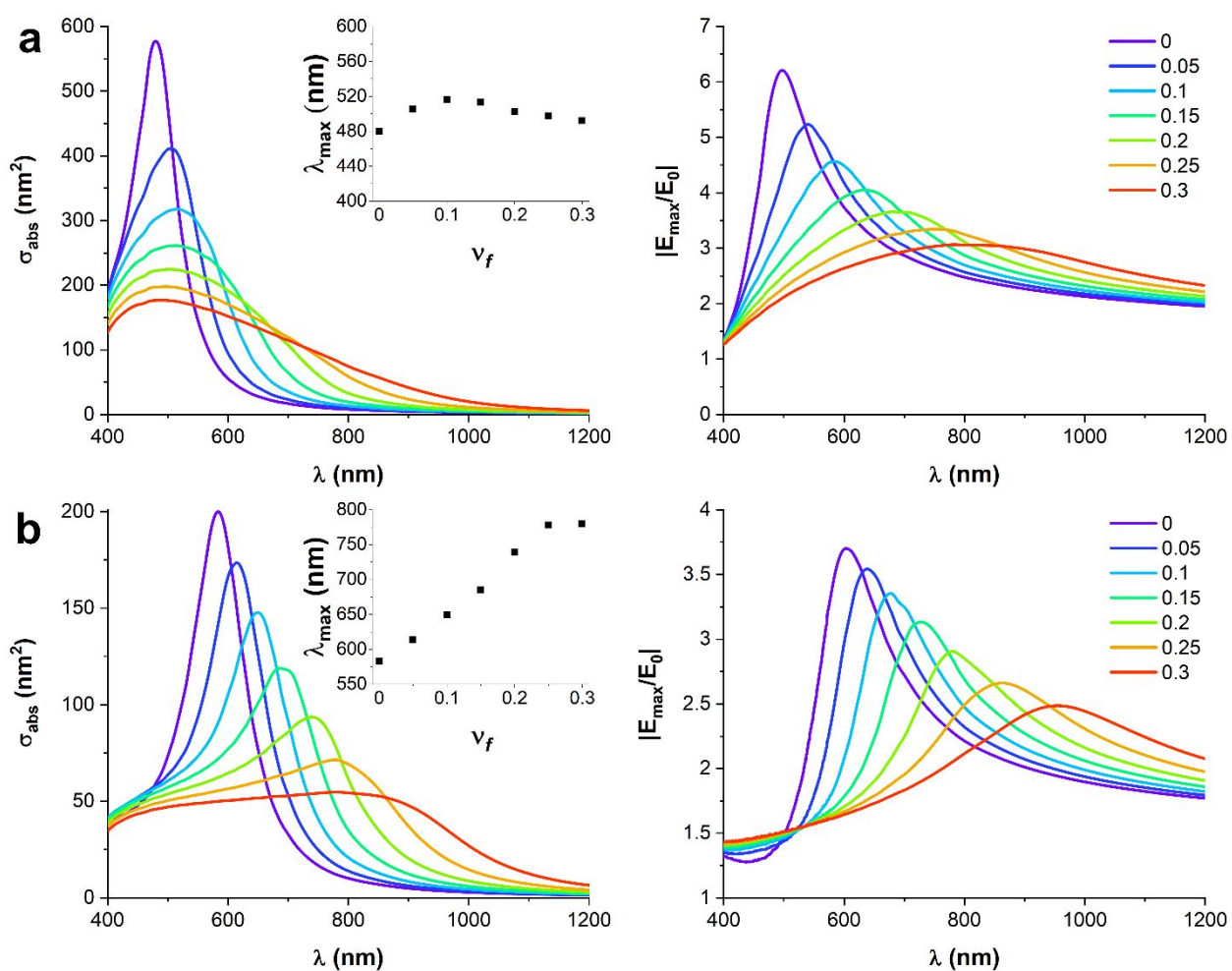


Figure S5. Calculated plasmonic absorption (left) and field-enhancement (right) properties of ZrN-based nanoparticles. **(a)** ZrN/ZrO₂ core structure; **(b)** ZrN/ZrO₂@a-C core@shell structure. Inset shows dependence of peak wavelength vs volume fraction v_f of oxide inclusions.

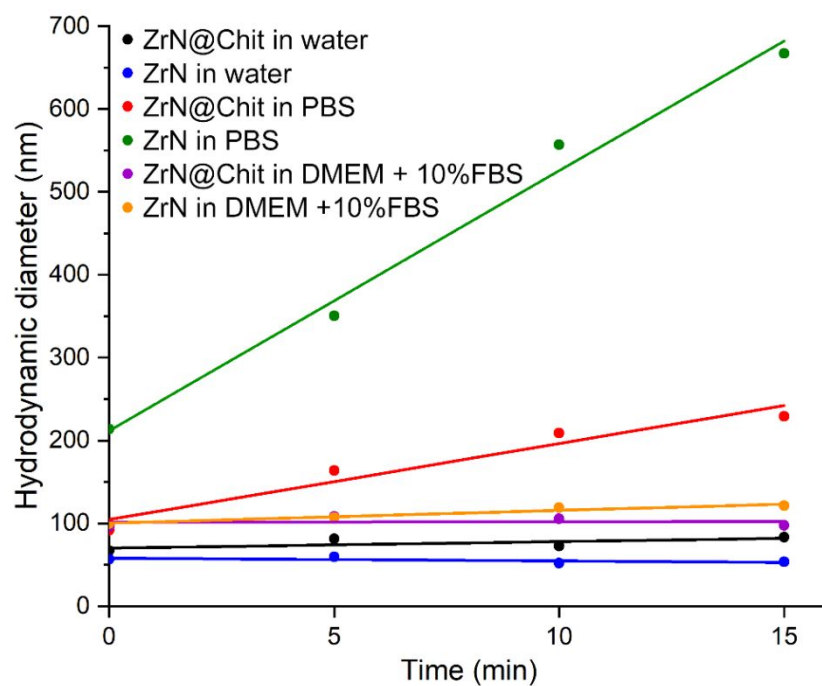


Figure S6. Stability evaluation of ZrN and ZrN@Chit nanoparticles in different buffers.

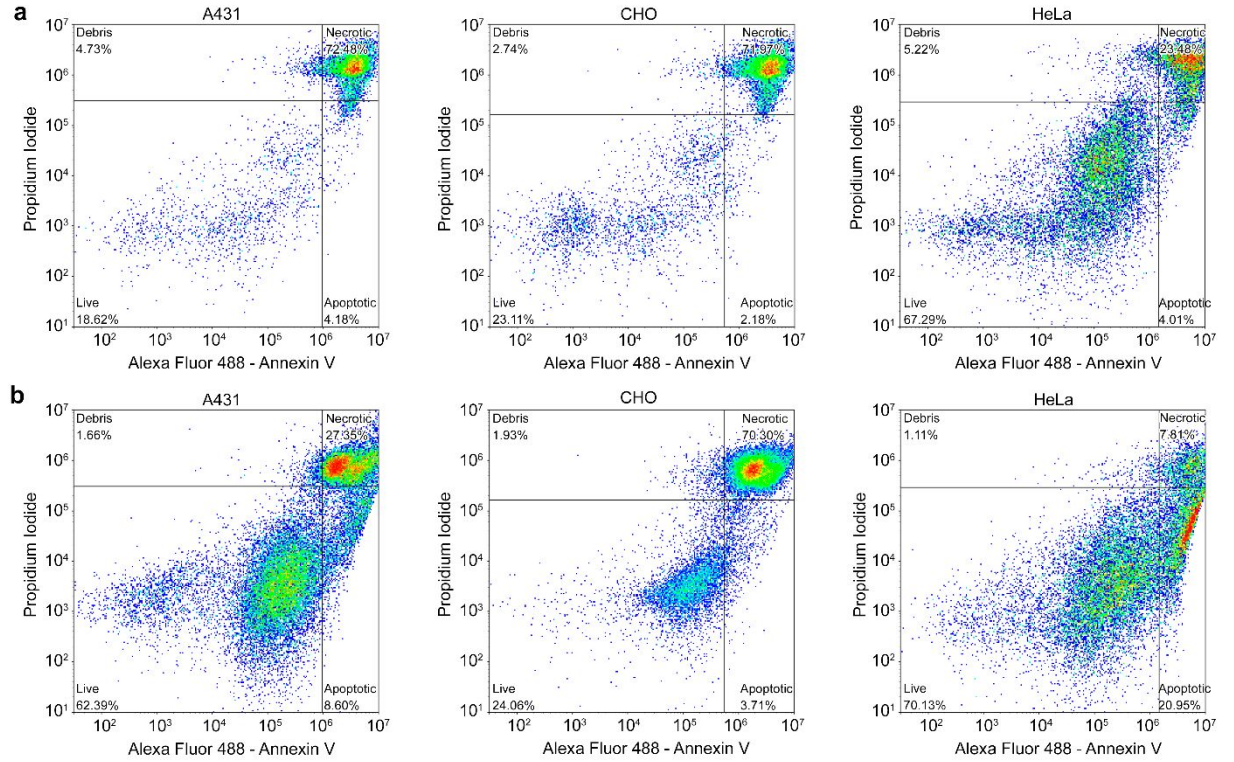


Figure S7. Cytometry investigation of quantity of apoptotic and necrotic cells after the heating at 75 °C for 10 min (a) and after their incubation with 1 μ M staurosporine for 4 h (b).

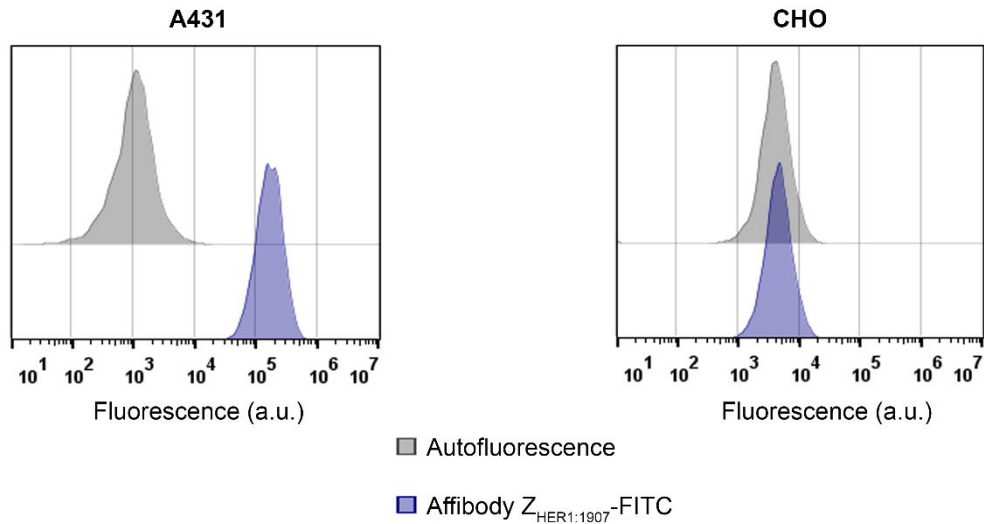


Figure S8. Labeling of EGFR-overexpressing A431 cells and EGFR-negative CHO cells with Z_{HER1:1907} affibody.

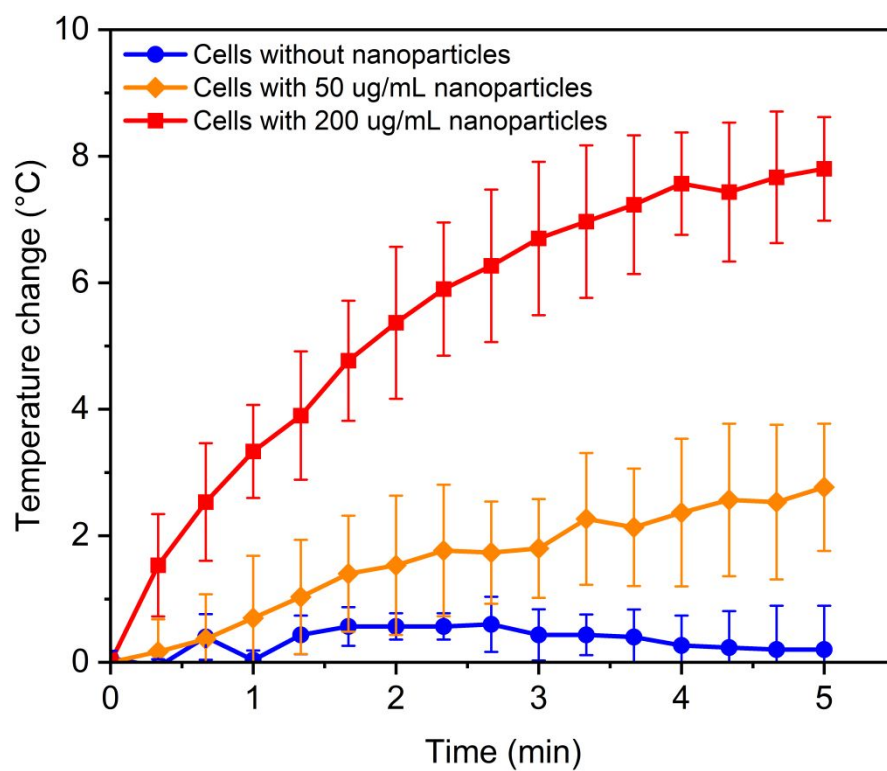


Figure S9. Temperature measurements of cell suspensions labeled with ZrN-based NPs at different concentrations during irradiation with 808 nm laser. Laser power is 1200 mW.

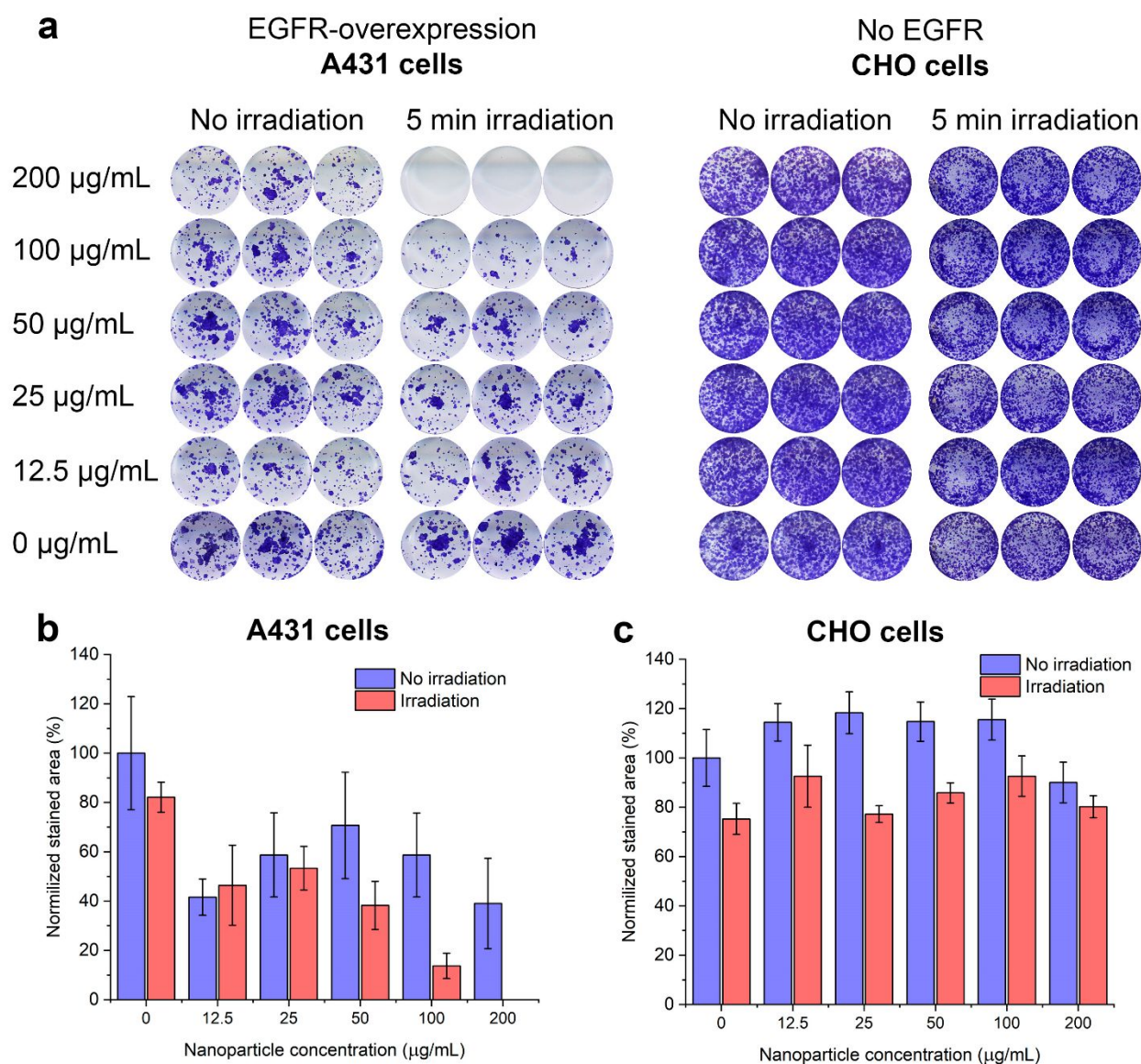


Figure S10. Clonogenic assay assessing $\text{ZrN@Chit*Z}_{\text{HER1:1907}}$ NPs cytotoxicity for A431 and CHO cells: **(a)** images of formed colonies and **(b,c)** quantitative evaluation of stained area surface. Cells were incubated with $\text{ZrN@Chit*Z}_{\text{HER1:1907}}$ NPs at different concentrations, irradiated with 805-nm light, and seeded on culture plates to get colonies being formed. Data was normalized to the cells grown without nanoparticles and laser irradiation.

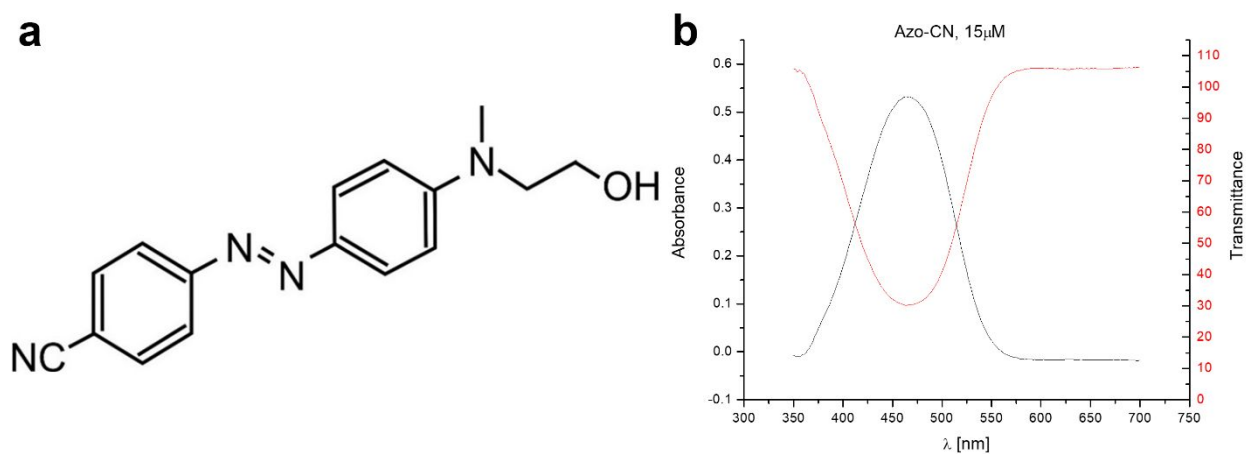


Figure S11. Azo-CN-OH molecule and its optical characteristics. **(a)** Azo-CN-OH structure. **(b)** Azo-CN-OH absorption and transmission spectra.

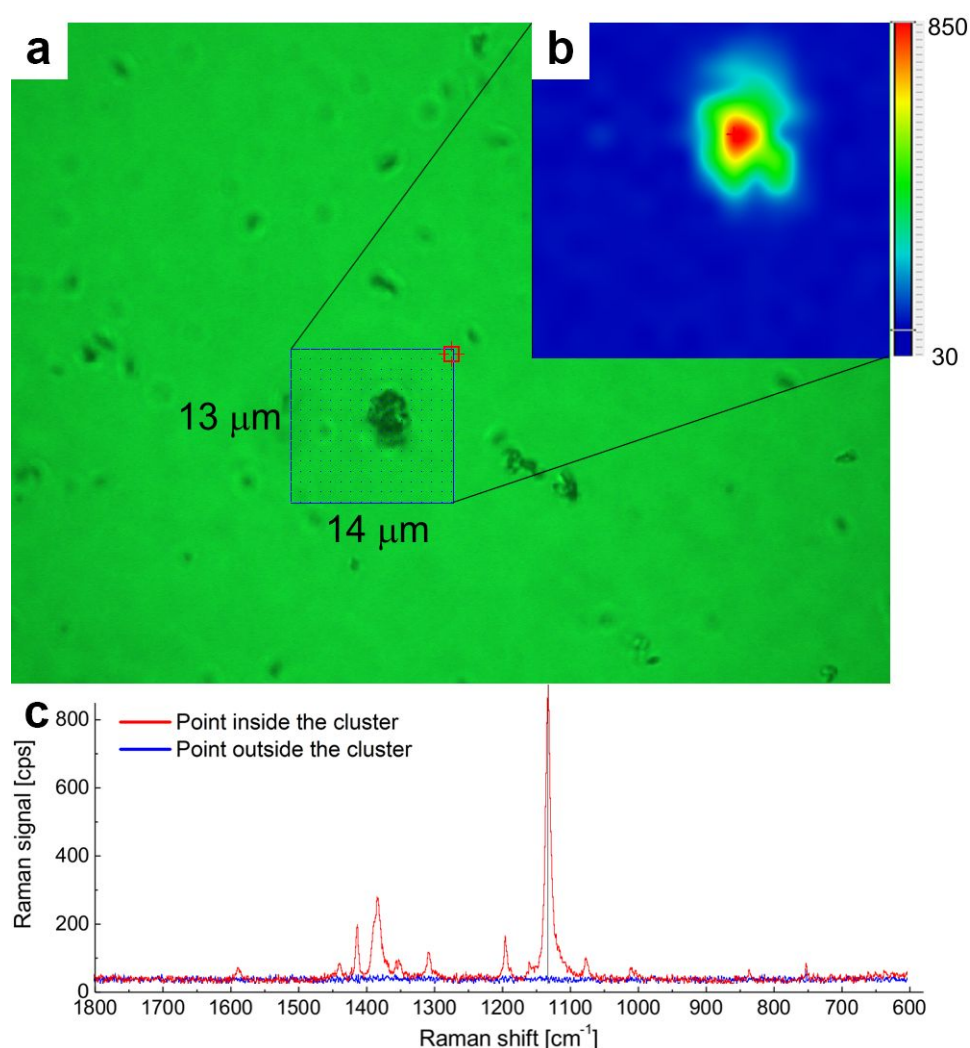


Figure S12. Imaging of ZrN/Azo-CN-OH conjugate cluster on the glass substrate in water. Scanning area was $14 \times 13 \mu\text{m}^2$. Accumulation time per point is 500 ms; scanning step is $1 \mu\text{m}$. Laser power is 5 mW@785nm. **(a)** Transmission image, scanned area is outlined by blue rectangular. **(b)** Raman map at 1133 cm^{-1} . **(c)** Raman spectra, corresponding map points with maximum (red) and minimum (blue) Intensities in the imaging **(b)**.

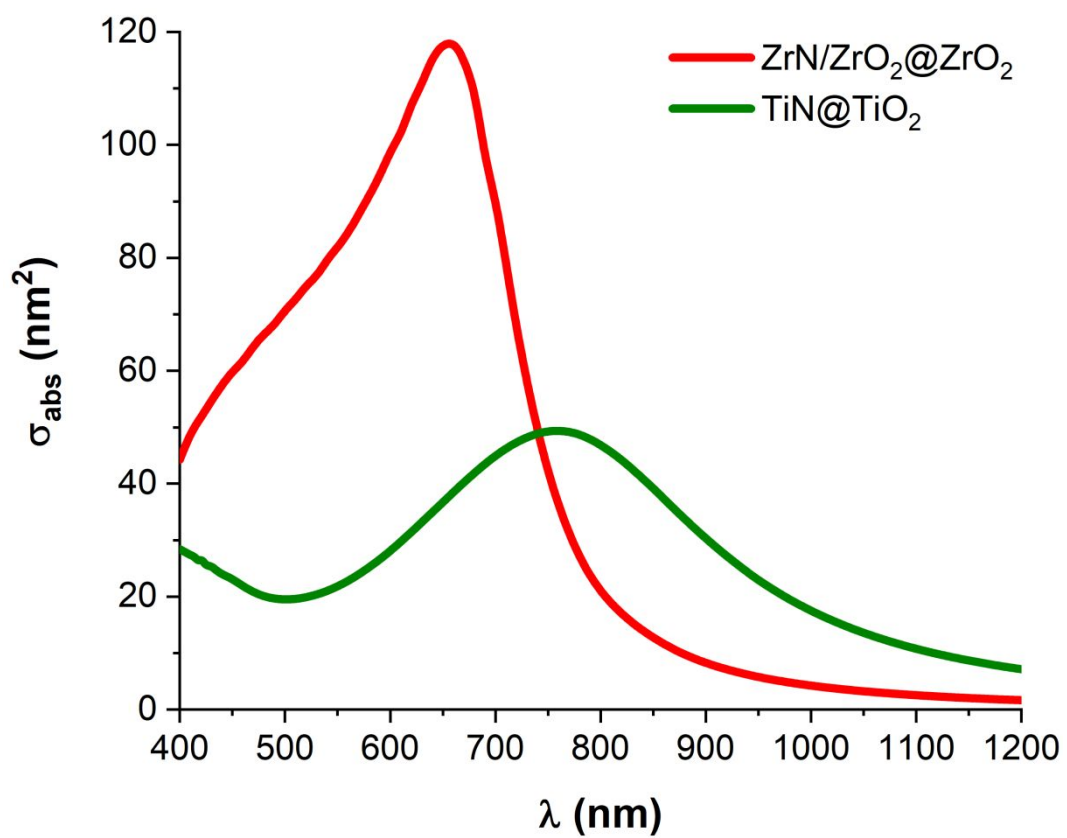


Figure S13. Calculated absorption cross-section of 20 nm ZrN and TiN-based core@shell nanoparticles.